

Description for Shabd Psycholinguistic Database

There are four lists of words made available as part of the database.

If you use any of the word lists in the database in your experiments, please cite us as:

Verma et al. (2020) Shabd: A Psycholinguistic Database for Hindi. (submitted).

List of 34k words.

List of 96k words.

List 2.3M words.

List 2.5M most frequent bigrams.

The following columns are present in the file with 34k words respectively. The columns are as follows:

- Word
- Frequency: Word Frequency as observed in Shabd Corpus.
- Frequency Per Million: Frequency of the word per million words observed in Shabd Corpus.
- Zipf: Frequency of the word as expressed in zipf-values, as observed in Shabd Corpus. (as defined by van Heuven et al. 2014)
- AkshCount: Total number of *aksharas*, i.e. orthographic units in *Devanagari* script.
- MatraCount: Total number of matras, i.e. the number of ligatures in the word.
- Total Length: Akshara Count + Matra Count
- PosTag1-PosTagFr1 – PosTag8-PosTagFr8: List of 8 most dominant pos-tags for the given word, with their frequencies.

- OLD 20: Orthographic Levenshtein Distance, measurement of orthographic similarity to the closest 20 words as defined by Yarkoni et al. (2008).
- CD: number of documents in which the word appears, measurement of contextual diversity as defined by Adelman et al. (2006).
- Log CD: log10 of CD value.
- Prosodic Label (PLSB): Prosodic Label of each word with stress markers which may be useful to speech and phonology researchers.
- I-level Syllabification and SyllCount (based on I-level syllabification): Syllabification which takes into account the inherent schwa vowel in Hindi *aksharas*, and syllable count as per this syllabification scheme.
- Phoneme_Level_ASCII Syllables and Syllcount based on the same: This syllabification scheme is the more commonly used syllabification scheme, where in the scheme takes into account the inherent schwa deletion, which is the case in common Hindi language use. For most purposes this measure of syllabification may be preferred.
- IPA_Equivalent and IPA Equivalent Phoneme count: Phoneme separation based on stress patterns and phoneme count based on these. May be more useful to phonology researchers.
- Phonemes_Level_IPA and Phonemes_Level_IPA Phoneme count: Phoneme separation using the more common IPA notation and phoneme count based on the same. This might be the commonly preferred phoneme counts for most researchers.
- Frequency_EMILLE: Frequency of the word as observed in the EMILLE Corpus.
- FreqPerMillion_EMILLE: Frequency of the word per million words as observed in the EMILLE Corpus.
- Zipf_EMILLE: Zipf value as observed in the EMILLE corpus.
- WordlexBlog Freq: Frequency of the word as observed in the Wordlex Blogs corpus.

- Wordlex_Blog Frequency per million: Frequency of the word per million words as observed in the Wordlex Blog Corpus.
- WordlexBlog Zipf Value: Frequency of the word expressed in Zipf Values as observed in the Wordlex Blog Corpus.
- WordlexBlogCD: number of documents the word was observed in the Wordlex Blog Corpus.
- WordlexBlogCDPc: percentage of documents the word was observed in the Wordlex Blog Corpus
- WordlexNews Freq: Frequency of the word as observed in Wordlex News Corpus
- WordlexNews Frpm: Frequency of the word per million words as observed in the Wordlex News Corpus.
- WordlexNews ZipfValue: Frequency of the word expressed in Zipf Values as observed in the Wordlex News corpus.
- WordlexNewsCD: number of documents a word was observed in the Wordlex News Corpus.
- WordlexNewsCDPc: percentage of documents the word was observed in the Wordlex News Corpus.

The list of 96k words has most of the information that is available in the 34k list. However, the following information is **not** available:

- OLD 20

The list of 2.3 M words only has the following columns:

- Word

- Frequency
- Length
- Zipf Frequency
- Frequency EMILLE
- Zipf EMILLE
- WordlexBlog Freq
- Wordlex Blog FreqPM
- Wordlex Blog Zipf Value
- Wordlex Blog CD
- Wordlex Blog CDPc
- Wordlex News Freq
- Wordlex News FreqPM
- Wordlex News Zipf
- Wordlex News CD
- Wordlex News CDPc